ELECTRICITY

1. <u>Switch</u> :- A switch makes a conducting link b/w cell and bulb

2. <u>Electric Circuit</u> :- A continous and ~~co~~ closed path of electric current

3. <u>Current</u> :- Expressed by amnt. of charge flowing through a particular area in unit time.
   ↳ rate of flow of $\underbrace{\text{electric charges}}_{\text{constituted by electrons}}$.

4. $I = \frac{Q}{t}$

   $Q \rightarrow$ S.I. = coulomb (C)

   $1C = 6 \times 10^{18}$ electrons

   charge possessed by $1e^- = 1.6 \times 10^{-19} C$

5. Electric Potential

   Potential Difference - Work done to move a unit charge from one point to another.

   ↳ Chemical activity within a cell generates the potential difference across the terminals of the cell even when no current is drawn from it.

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$$V = \frac{\text{Work done}}{Q(\text{Charge})}$$

→ S.I. unit = volt (V)

1 V is the potential diff. b/w 2 pts in a current carrying conductor when 1 joule of work is done to move 1 C charge from 1 pt. to another.

6. Ohm's Law:- The potential diff. across the ends of a conductor is directly proportional to the ~~ter~~ current flowing through it at a given temp.

$$V \propto I$$

$$\frac{V}{I} = R$$

R (Resistance):- property of a conductor to resist the flow of charges

Variable resistance:- component used to change regulate current w/o changing voltage

rheostat

7. $R = \rho \frac{l}{A}$

ρ (rho): constant of proportionality
- called electrical resistivity
Ωm

★ Metals & alloys: $10^{-8}\ \Omega m$ to $10^{-6}\ \Omega m$
Insulators: 10^{12} to $10^{17}\ \Omega m$

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★ Resistivity of alloys is ↑ than that of its metals.
They don't burn easily at high temp.
So ~~com~~ used in elec. heating devices

8. $R_{series} = R_1 + R_2 + \dots$

$$R_{\parallel} = \frac{1}{R_{\parallel}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

9. Problems in series circuit
↳ impractical in ~~devices~~ cases where devices need widely diff. I values to operate properly.
↳ when 1 component fails, entire circuit breaks

10. HEATING EFFECT OF ELECTRIC CURRENT
↳ Heating effect of electric current:-
When a resistor is connected to a source, all of the energy supplied by source is dissipated in the form of heat.

Elec. energy ⟶ heat energy

$P = VI$
$H = VIt$
$= I^2Rt$

$$\left.\begin{array}{l} H \propto I^2 \\ H \propto R \\ H \propto t \end{array}\right\} \text{Joule's Law of Heating}$$

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11. Applications :-

1. To emit light in bulbs
   - Filament should get hot & emit light
   - bulbs filled w/ N_2 & Ar to prolong the life of ~~bu~~ filament
   - Strong metal w/ 1 M.P. such as tungsten

2. Electric laundry iron, electric toaster, oven, kettle

3. Fuse :-

12. Electric Power

$$P = VI = I^2R = \frac{V^2}{R}$$

$P \downarrow$ Watt

$$\underbrace{1\,kWh}_{\text{unit}} = 1000 \text{ wat} \times 3600 \text{ sec}$$
$$= 3.6 \times 10^6 \text{ Joule}$$

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